Lab 1: Packet analysis at application layer using Wireshark
SCSR1213 Network Communications
Universiti Teknologi Malaysia

Objective:

1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.

2. To introduce student with Wireshark software tool for packet analyzer.

3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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Section : 02

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PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures ("sniffs") messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

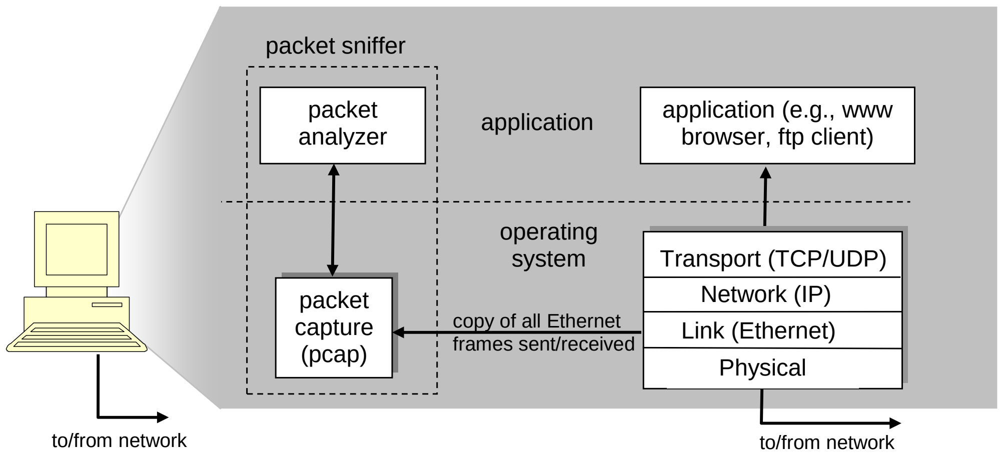


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must "understand" the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string "GET," "POST," or "HEAD".

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

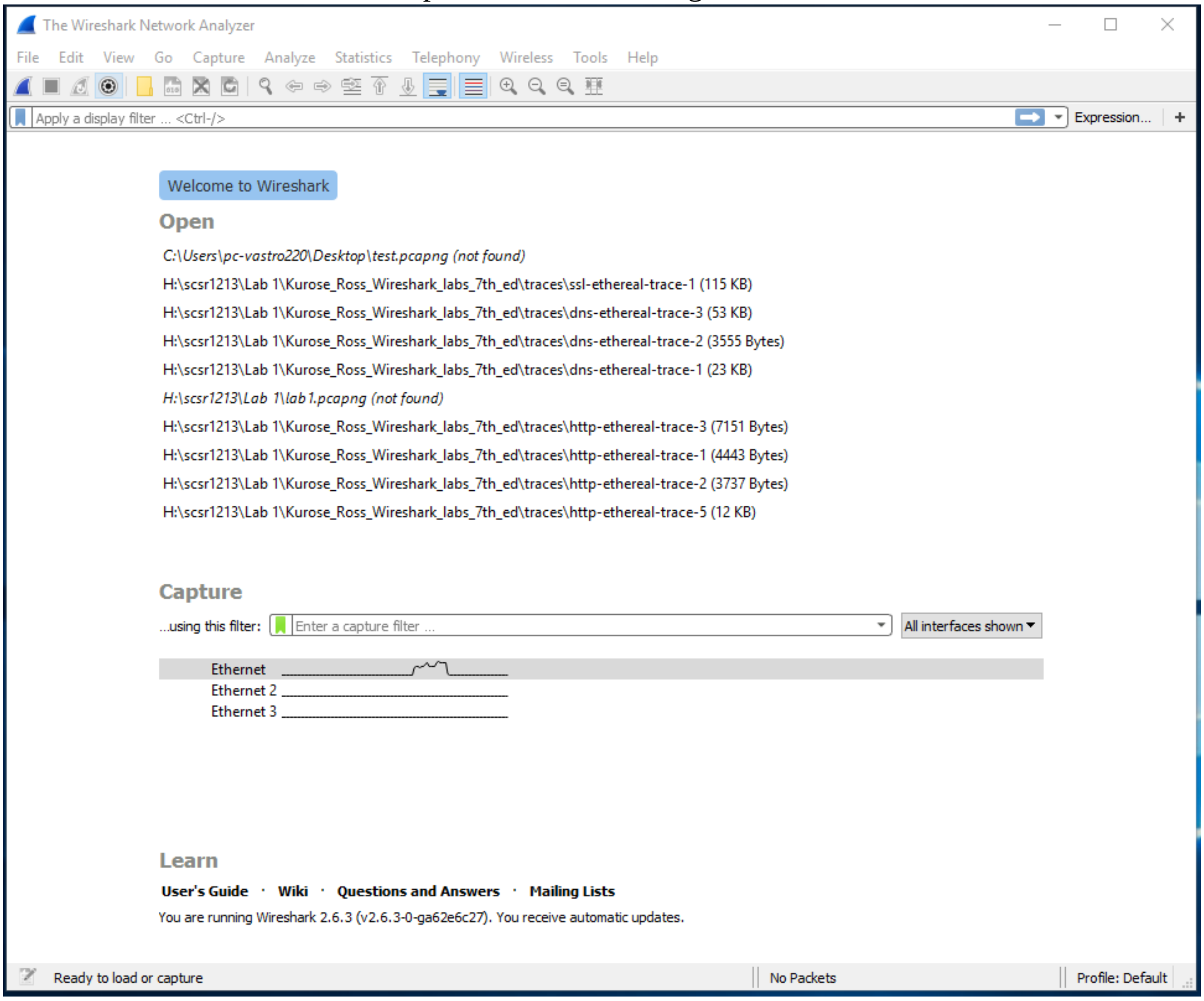


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

command menus

display filter specification

listing of captured packets

details of selected packet header

packet **content** in hexadecimal and ASCII

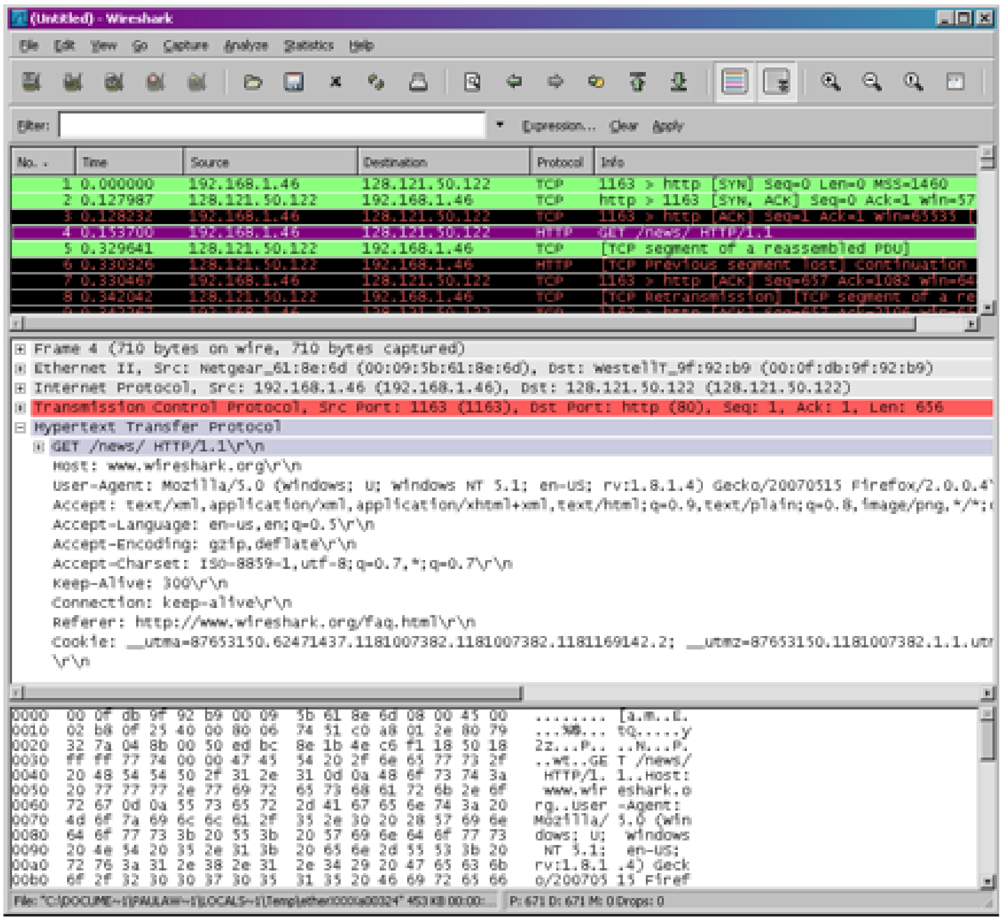


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field,** into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

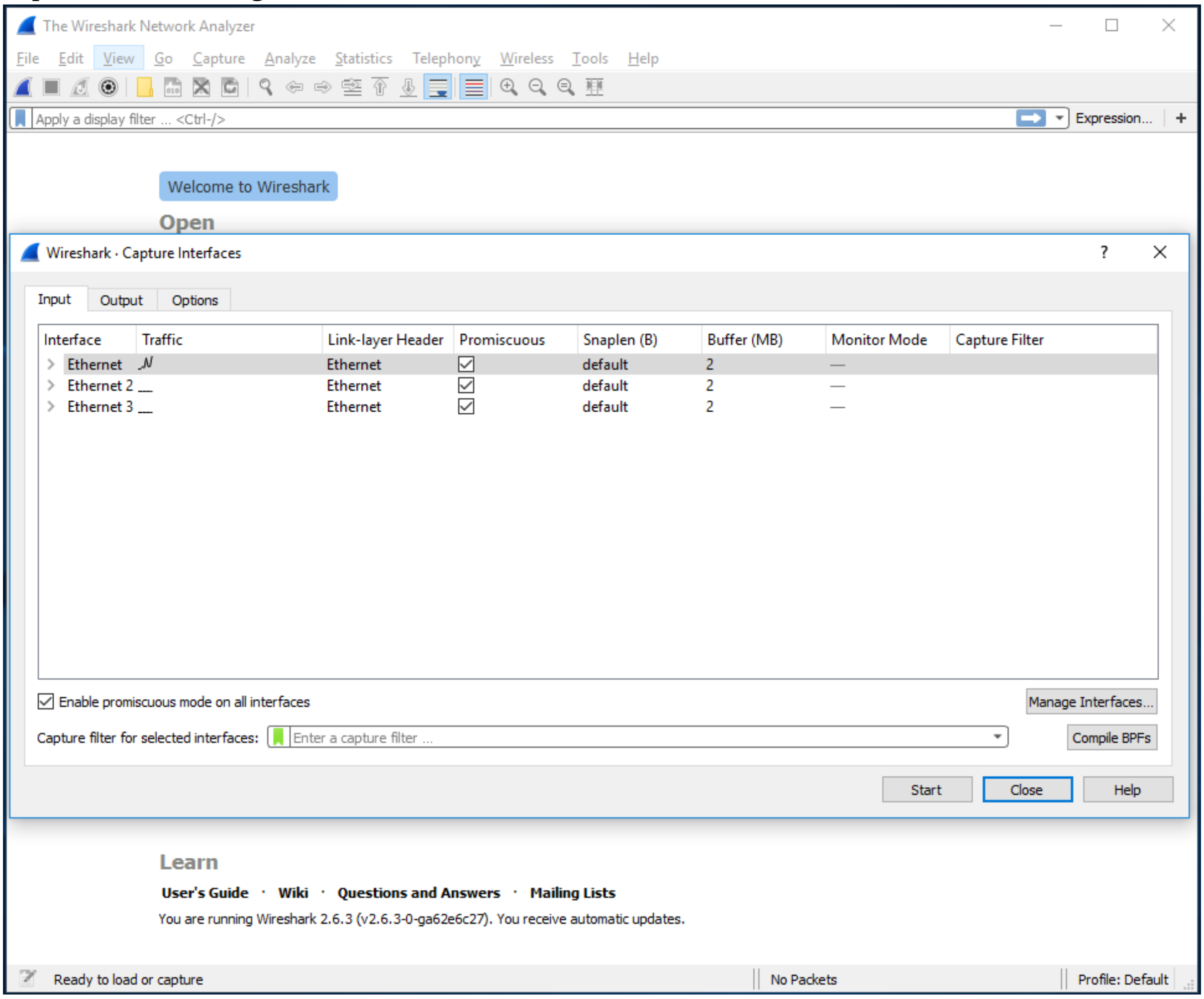


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

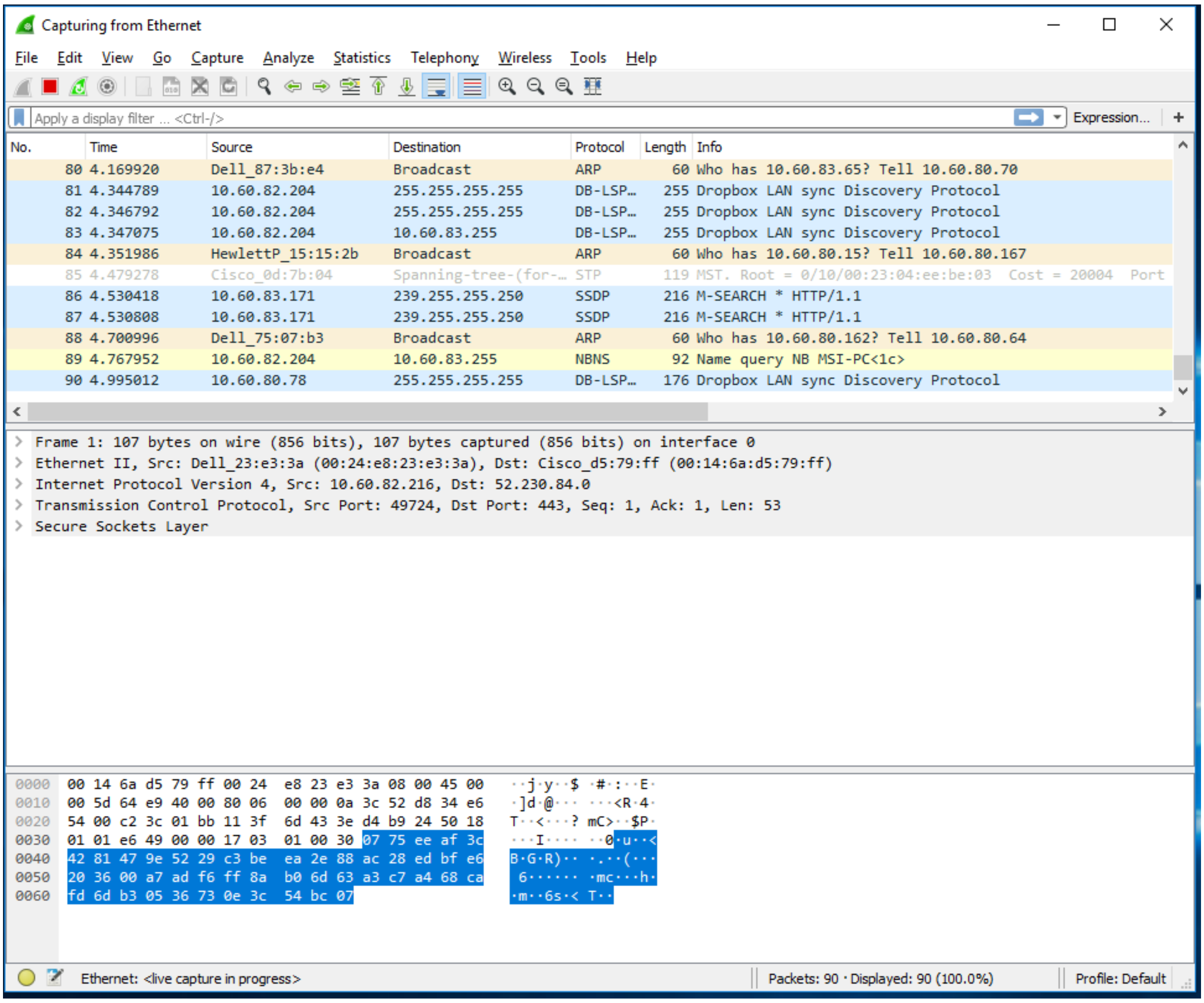


Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

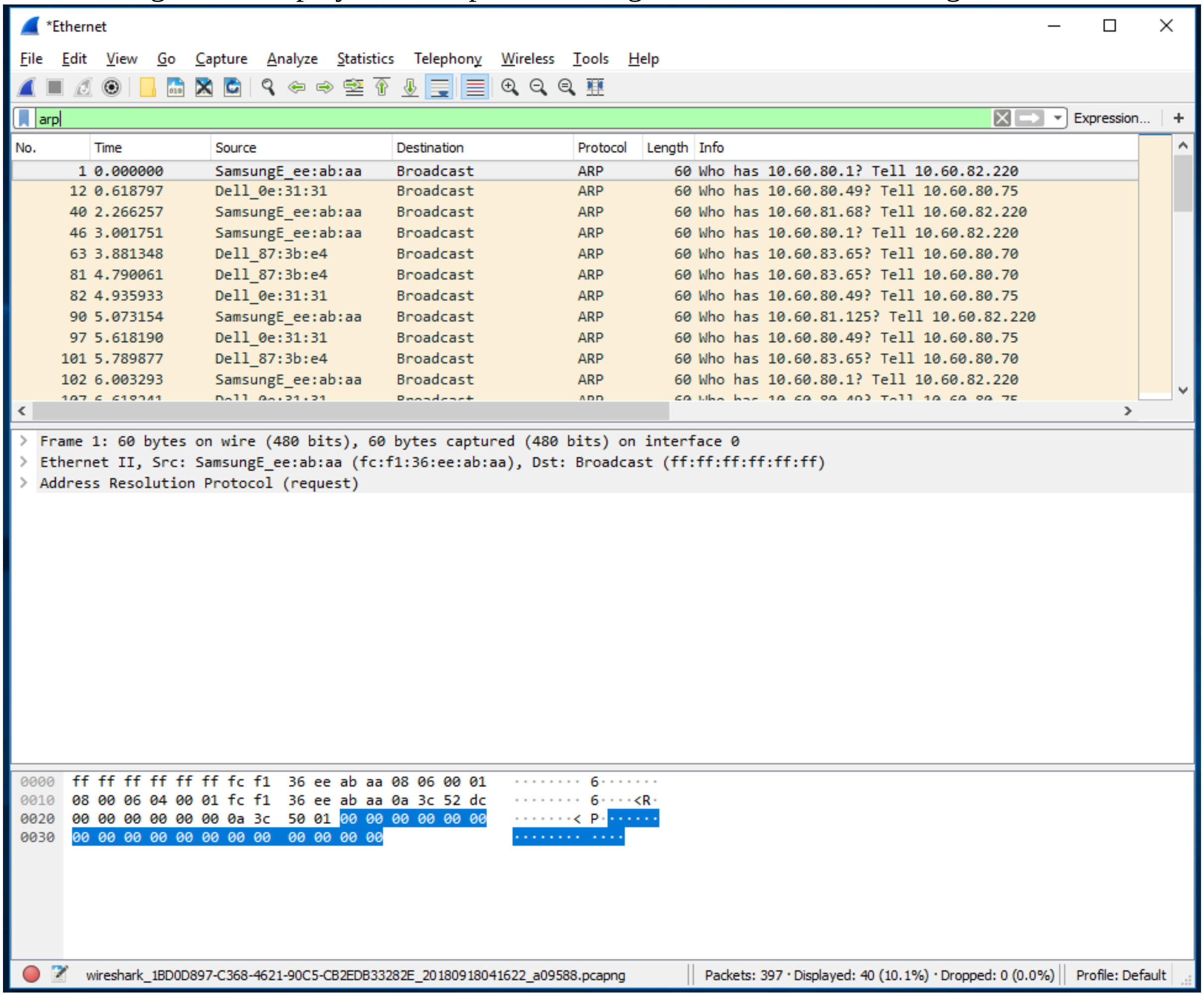


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

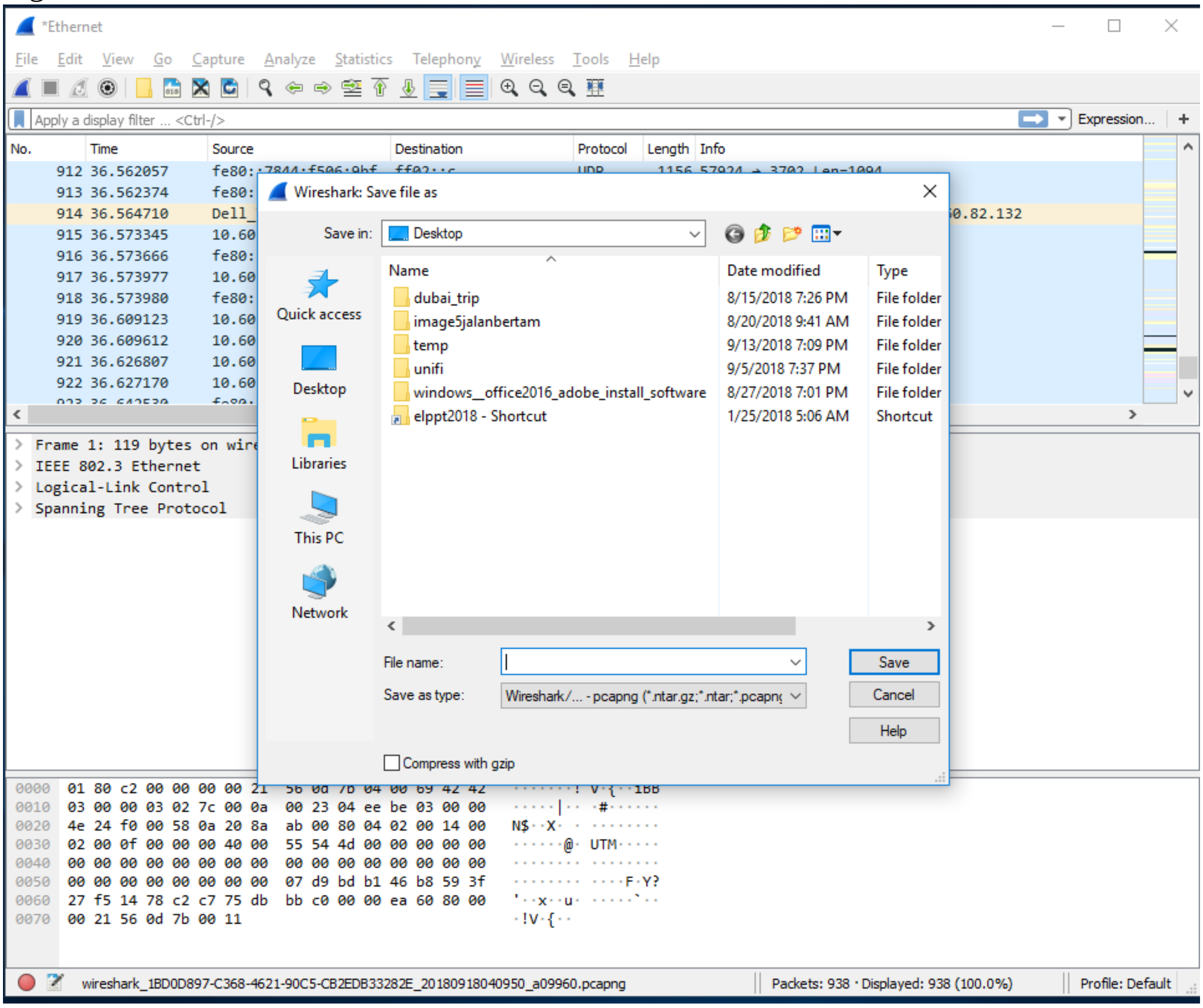


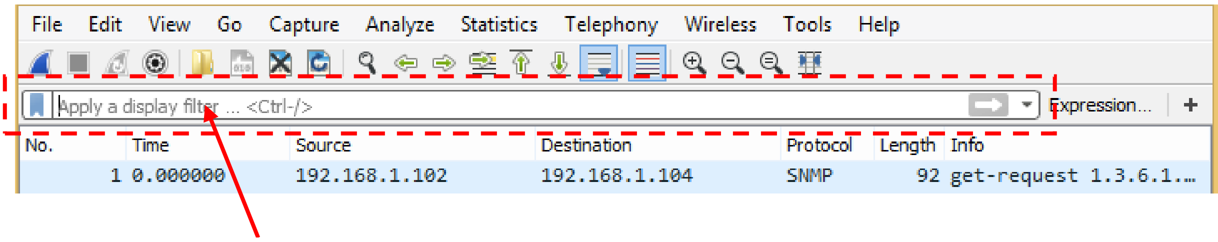
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter "**http**" (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



packet display filter

- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. What version of HTTP is the server running?

   - Version 1.1

2. What is the IP address of the client computer?

   - 192.168.102

3. What is the IP address of the gaia.cs.umass.edu server?

   - 129.119.245.12

4. How many bytes of content are being returned to client browser?

   - 73 bytes

5. What is the status code returned from the server to client browser?

   - 200 OK and 404 Not Found

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

    - No

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

    - Yes, the server return the contents of the file because of the status code 200 with a response phrase of OK.

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE:" header?

    - Yes there is an "IF-MODIFIED-SINCE:" line. The information is "Tue, 23 Sep 2003 05:35:00 GMT\r\n"

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

    - The status code is 304 with a response phrase of "Not Modified". The server did not return the contents of the file because the contents of the file has not changed since the last access and the status code indicates that there were no changes since then.

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

    - 3 HTTP GET request.

2. To which Internet addresses were these GET requests sent?

    - 192.119.245.12, 165.193.123.218 and 134.241.6.82

3. any bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?

    - Yes, the content is 3357 bytes

4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

    - 15642 bytes

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2

```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server:  ns1.utm.my
Address:  161.139.250.2

Non-authoritative answer:
Name:    yahoo.com
Addresses:  2001:4998:58:1836::11
          2001:4998:c:1023::4
          2001:4998:44:41d::4
          2001:4998:44:41d::3
          2001:4998:58:1836::10
          2001:4998:c:1023::5
          72.30.35.10
          98.137.246.7
          98.137.246.8
          98.138.219.231
          98.138.219.232
          72.30.35.9


C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server:  jb.utm.my
Address:  161.139.250.2

Non-authoritative answer:
yahoo.com       MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com       MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com       MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com       AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com       AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com       AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com       AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com       AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com       AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com       nameserver = ns5.yahoo.com
yahoo.com       nameserver = ns2.yahoo.com
yahoo.com       nameserver = ns4.yahoo.com
yahoo.com       nameserver = ns1.yahoo.com
yahoo.com       nameserver = ns3.yahoo.com
yahoo.com       internet address = 98.137.246.7
yahoo.com       internet address = 98.137.246.8
yahoo.com       internet address = 98.138.219.231
yahoo.com       internet address = 98.138.219.232
yahoo.com       internet address = 72.30.35.9
yahoo.com       internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result

```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server:  ns1.test.time.net.my
Address:  203.121.16.85

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4001:804::200e
          172.217.24.174


C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

```
C:\Windows\System32>nslookup www.microsoft.com
Server:  dns.tm.net.my
Address:  2001:e68::b:68
```

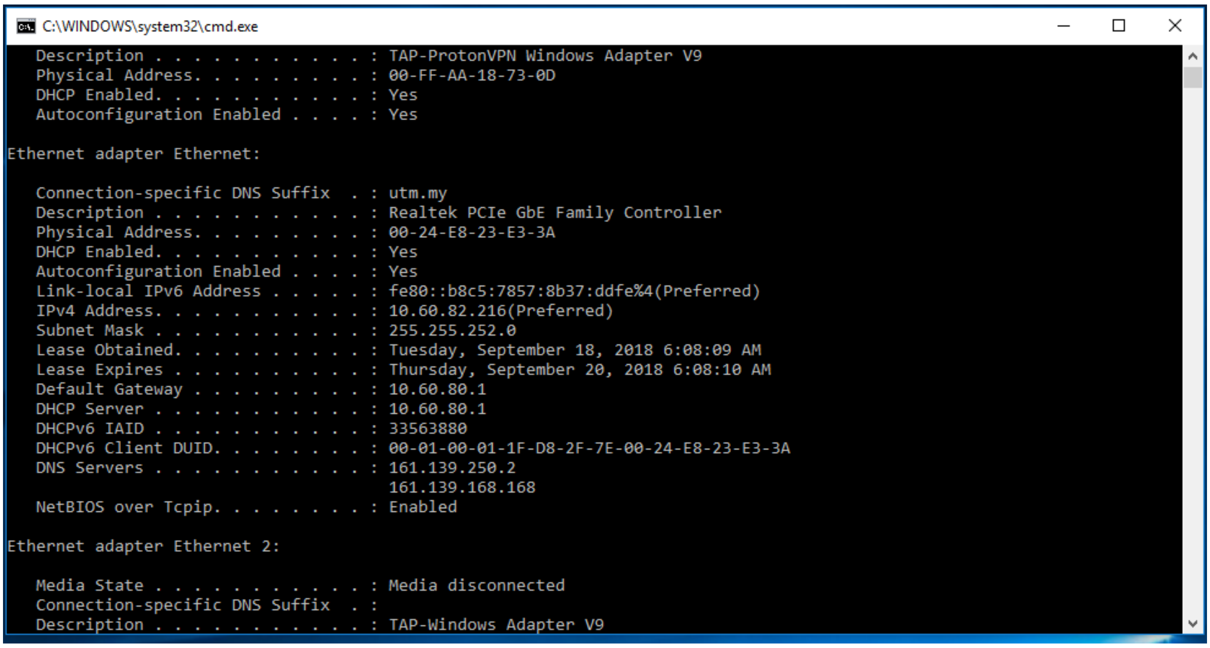
2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

```
Non-authoritative answer:
Name:    e13678.dscb.akamaiedge.net
Addresses:  2001:e68:2:38a::356e
          2001:e68:2:38d::356e
          2001:e68:2:388::356e
          2001:e68:2:38e::356e
          2001:e68:2:387::356e
          173.223.41.219
Aliases:  www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

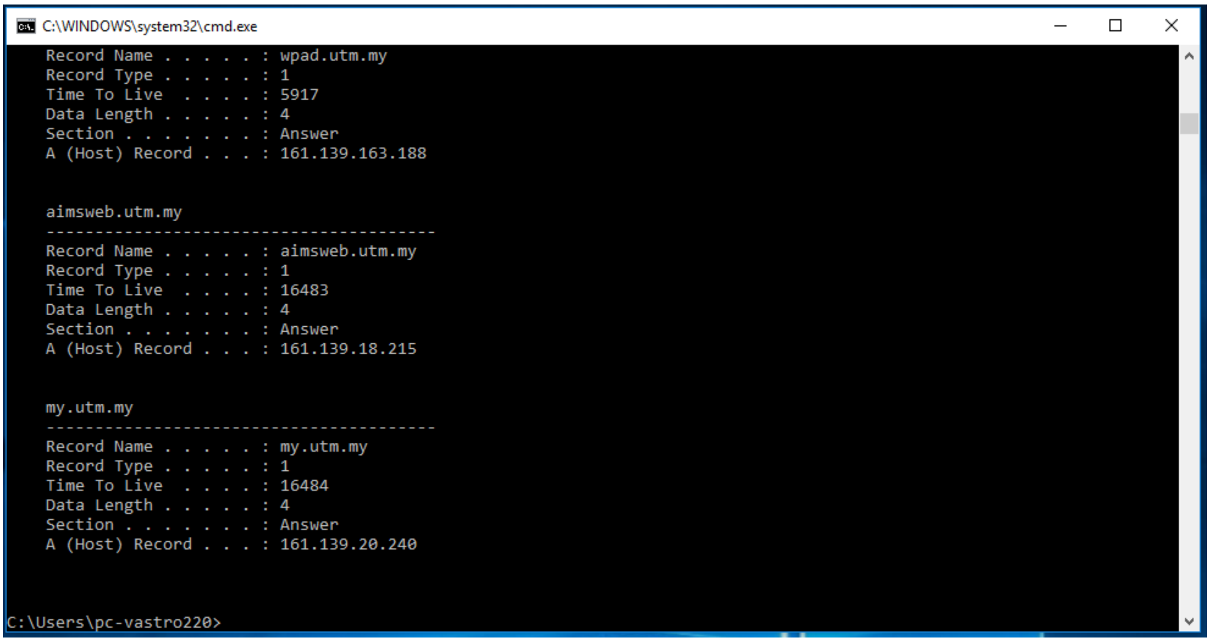
- Information about host, use the following command: ipconfig /all



```
C:\WINDOWS\system32\cmd.exe
   Description . . . . . . . . . . . : TAP-ProtonVPN Windows Adapter V9
   Physical Address. . . . . . . . . : 00-FF-AA-18-73-0D
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : utm.my
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : 00-24-E8-23-E3-3A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.60.82.216(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Lease Obtained. . . . . . . . . . : Tuesday, September 18, 2018 6:08:09 AM
   Lease Expires . . . . . . . . . . : Thursday, September 20, 2018 6:08:10 AM
   Default Gateway . . . . . . . . . : 10.60.80.1
   DHCP Server . . . . . . . . . . . : 10.60.80.1
   DHCPv6 IAID . . . . . . . . . . . : 33563880
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
   DNS Servers . . . . . . . . . . . : 161.139.250.2
                                       161.139.168.168
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
  Command: ipconfig /displaydns



```
C:\WINDOWS\system32\cmd.exe
    Record Name . . . . . : wpad.utm.my
    Record Type . . . . . : 1
    Time To Live  . . . . : 5917
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 161.139.163.188


    aimsweb.utm.my
    ----------------------------------------
    Record Name . . . . . : aimsweb.utm.my
    Record Type . . . . . : 1
    Time To Live  . . . . : 16483
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 161.139.18.215


    my.utm.my
    ----------------------------------------
    Record Name . . . . . : my.utm.my
    Record Type . . . . . : 1
    Time To Live  . . . . : 16484
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 161.139.20.240


C:\Users\pc-vastro220>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns

Figure C.5: ipconfig /flushdns result

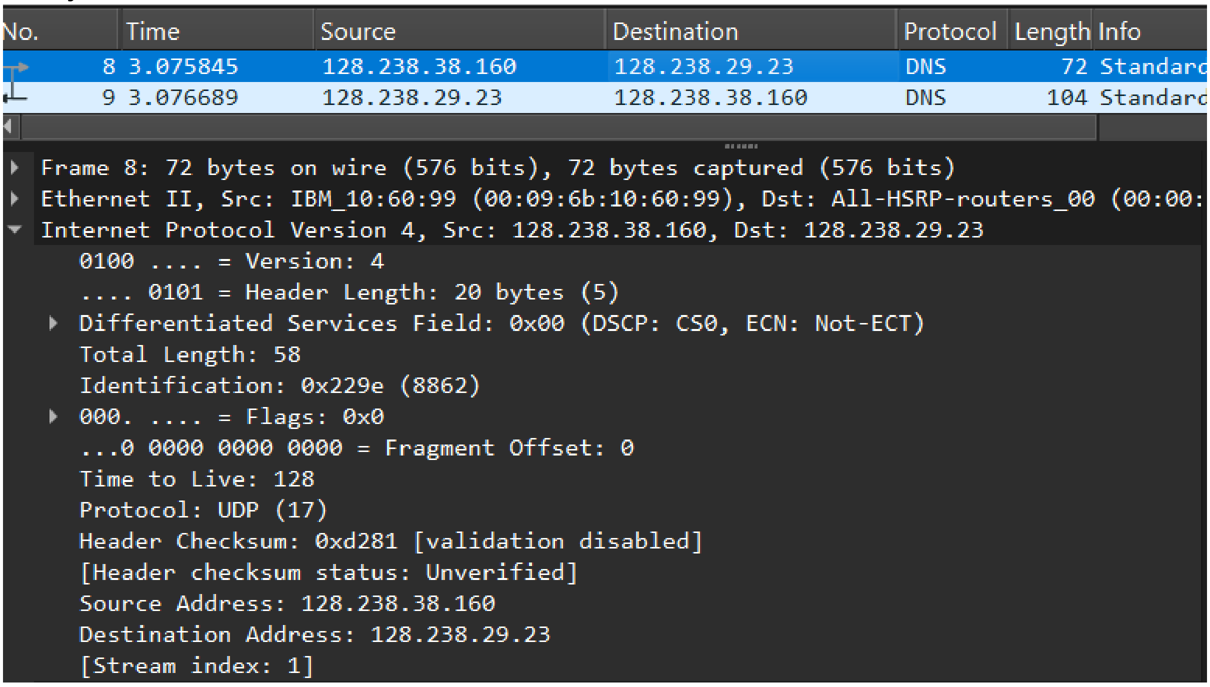
3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1. Answer the following questions.

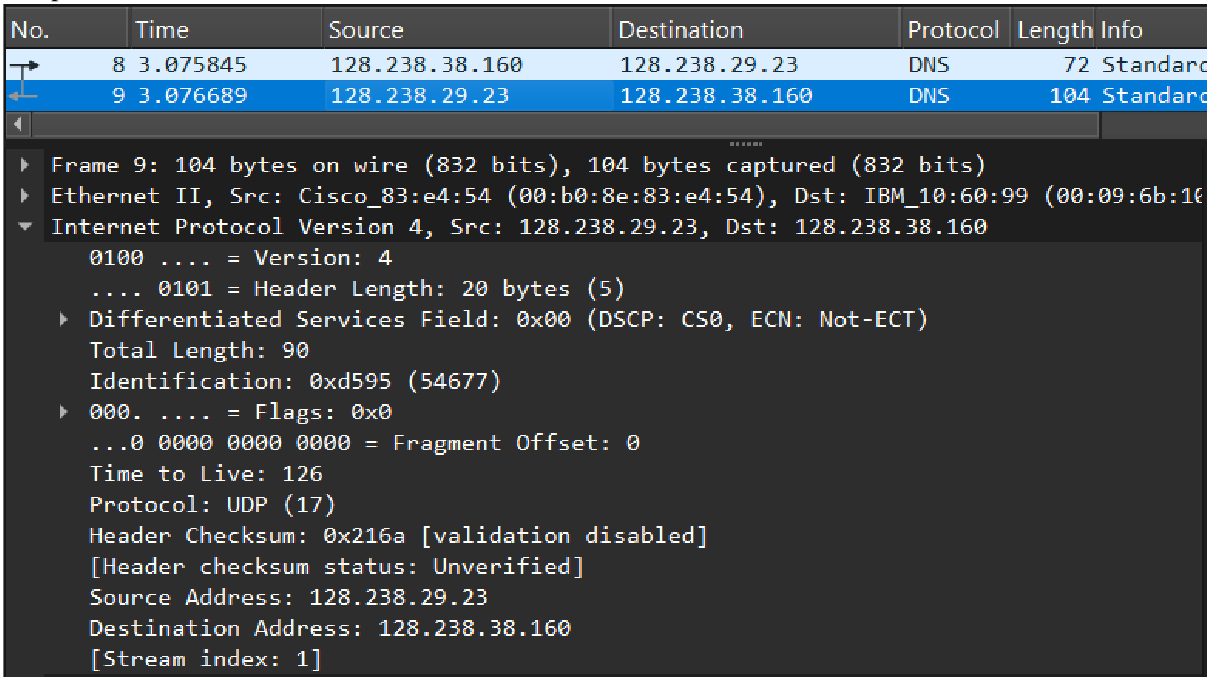
1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

   - The protocols for both query and response messages are UDP.

Query:

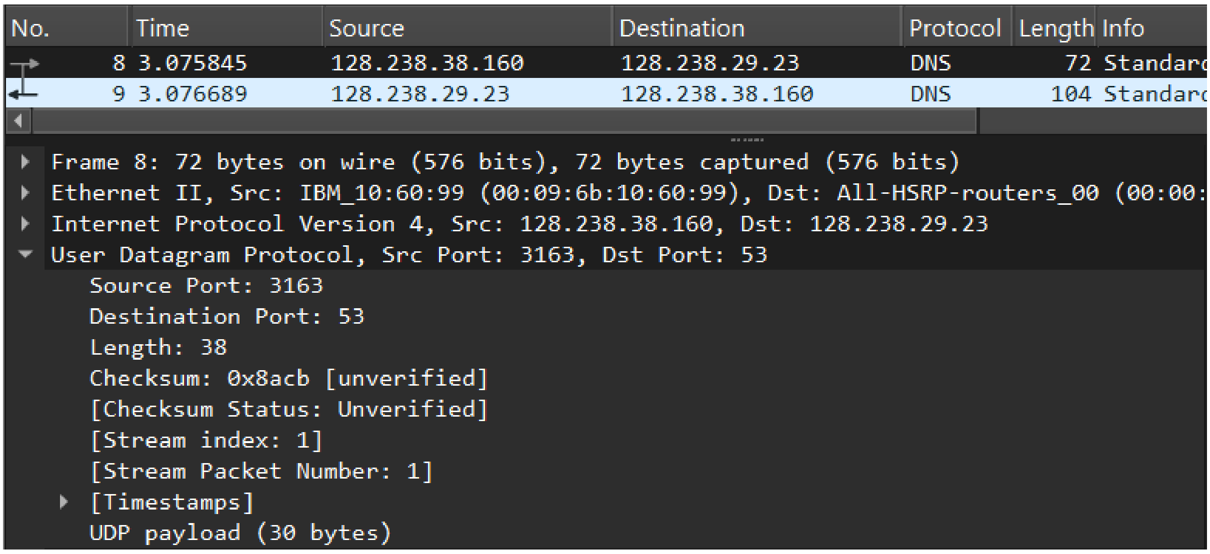


Response:

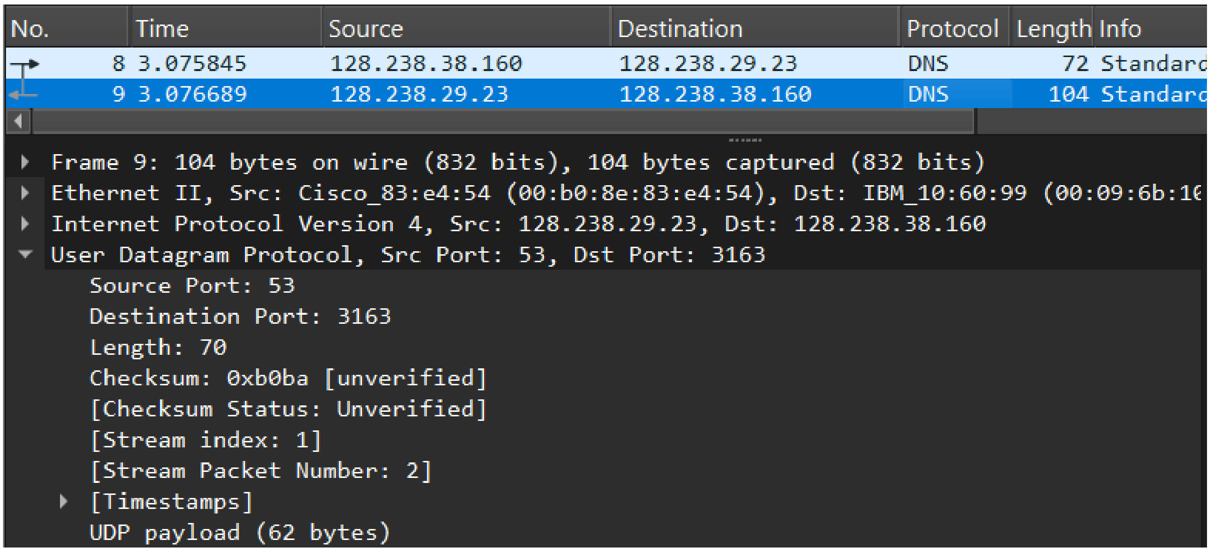


2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

- Destination port for the DNS query message: 53

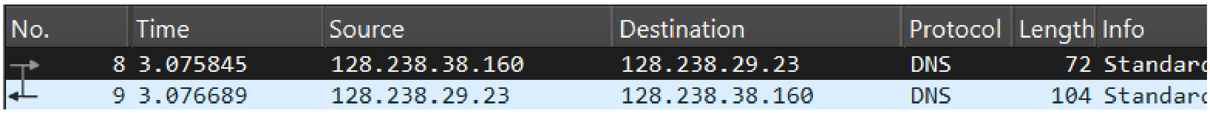


- Source port of DNS response message: 53



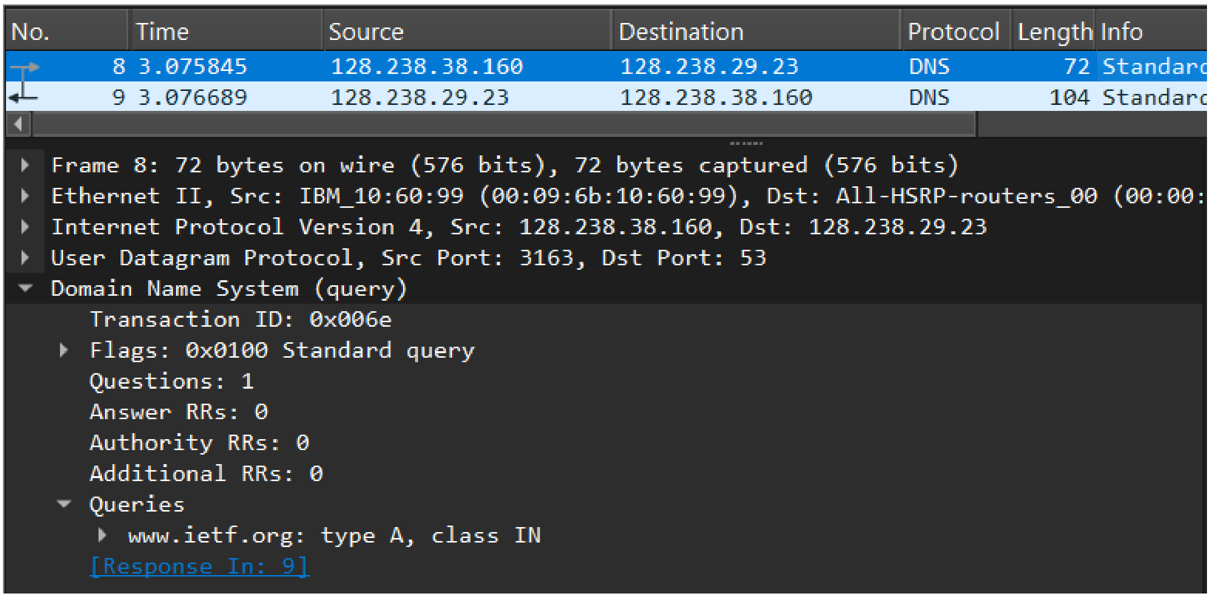
3. To what IP address is the DNS query message sent? Add screenshots in your answer.

- 128.238.29.23



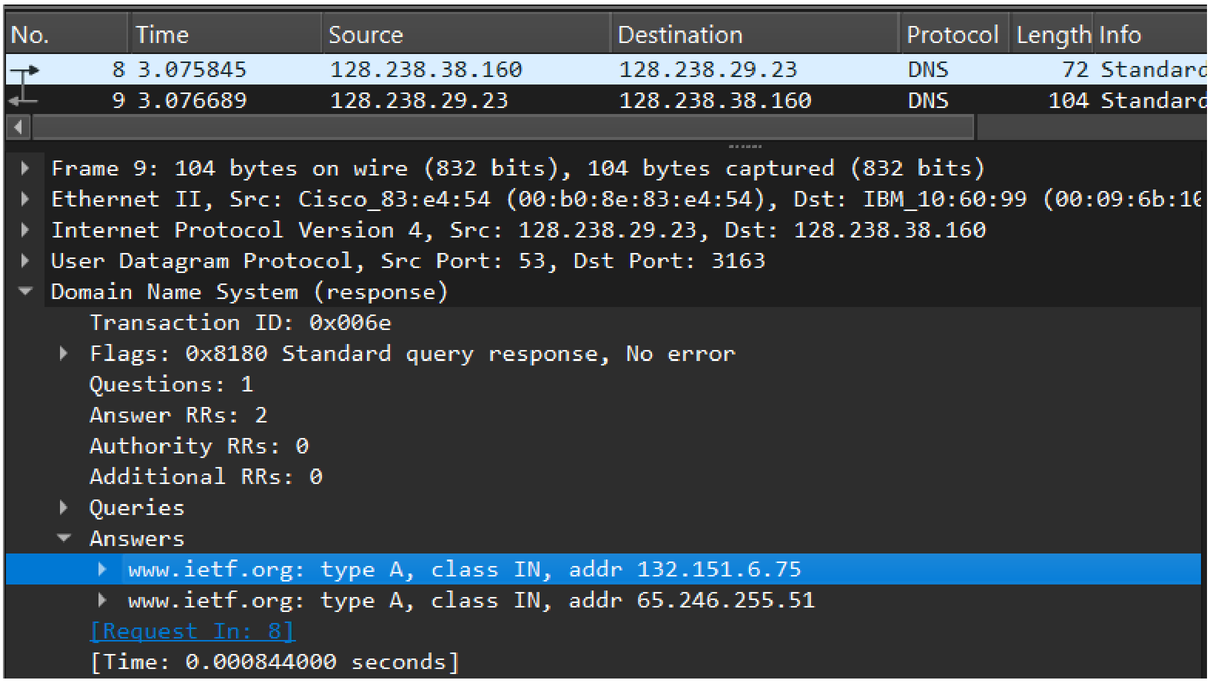
4. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"? Add screenshots in your answer.

- The type of DNS query message is type A
- The query message does not contain any answers

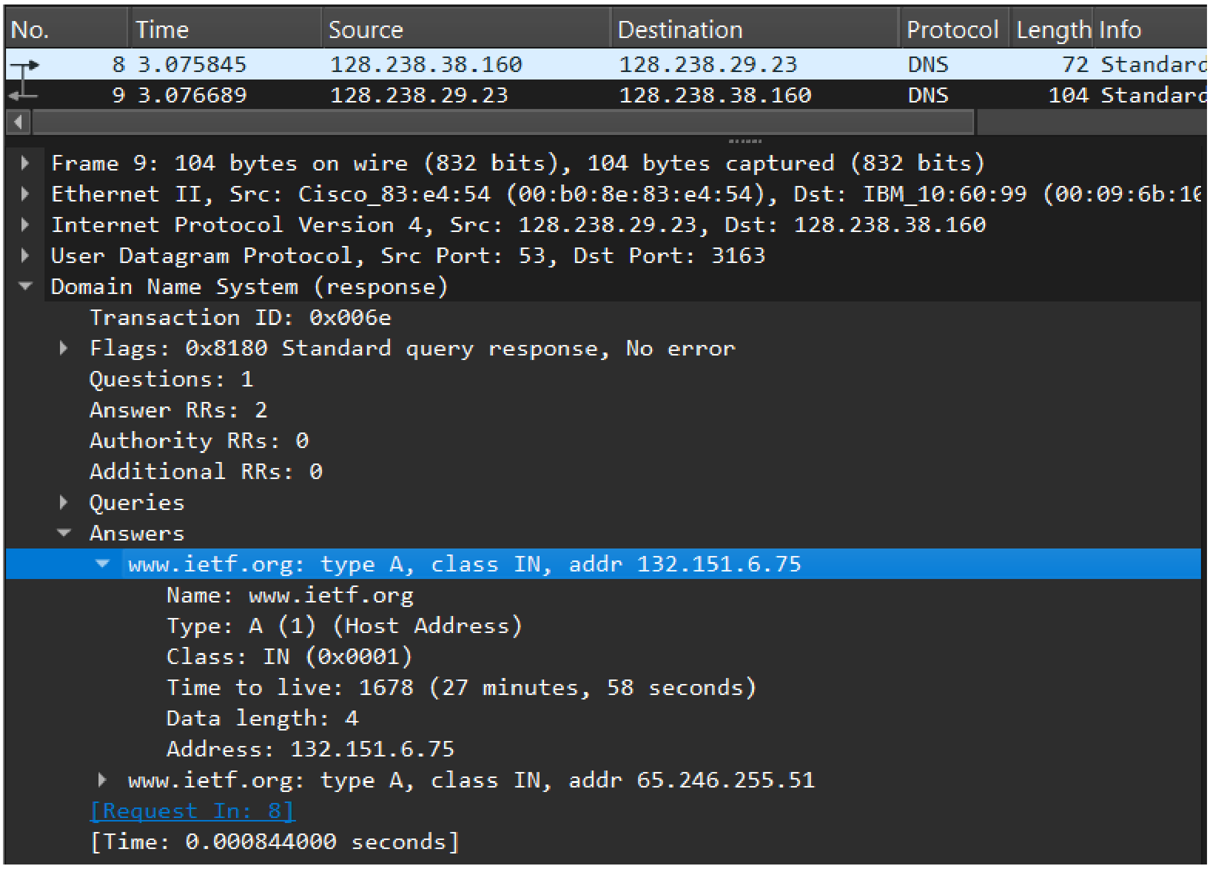


5. Examine the DNS response message. How many "answers" are provided? What do each of these answers contain? Add screenshots in your answer.

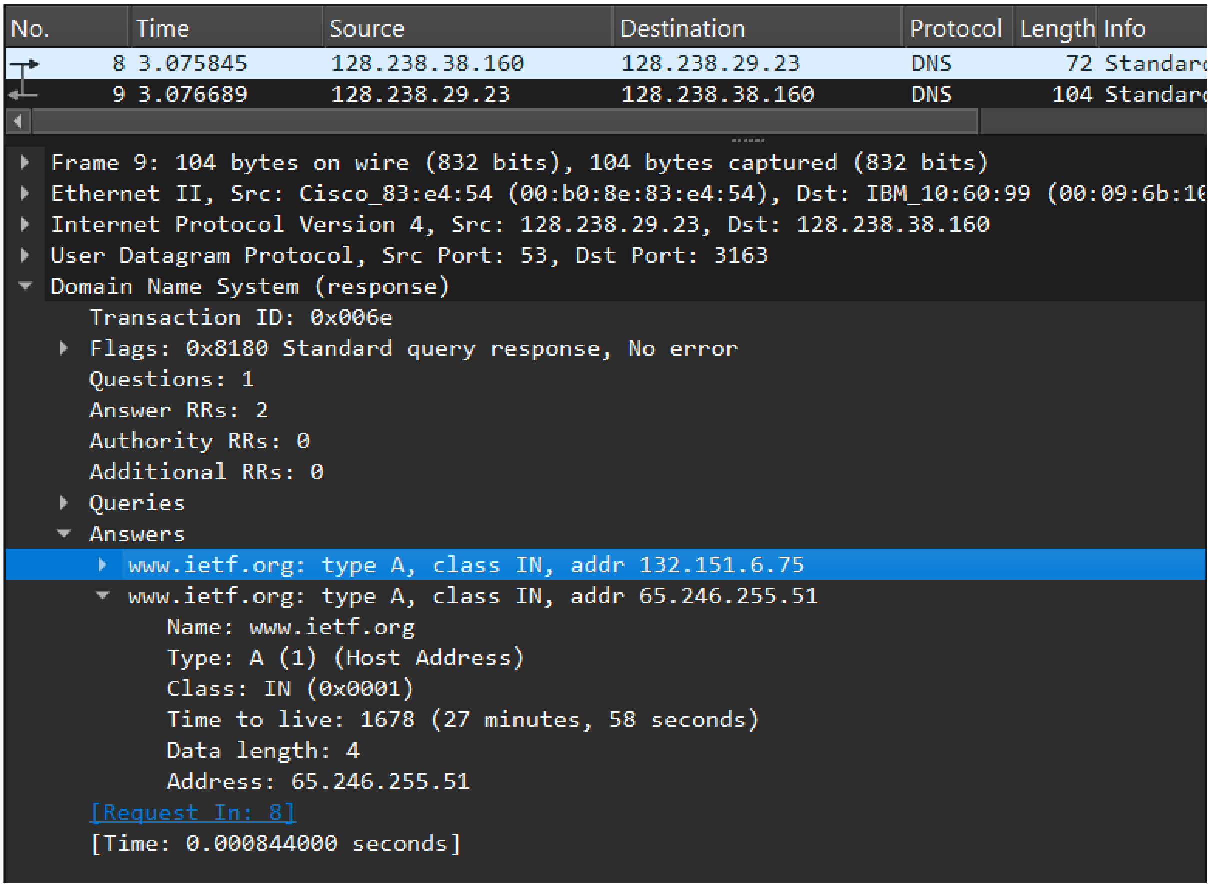
- Two answers are provided.



- Content of the 1st answer:

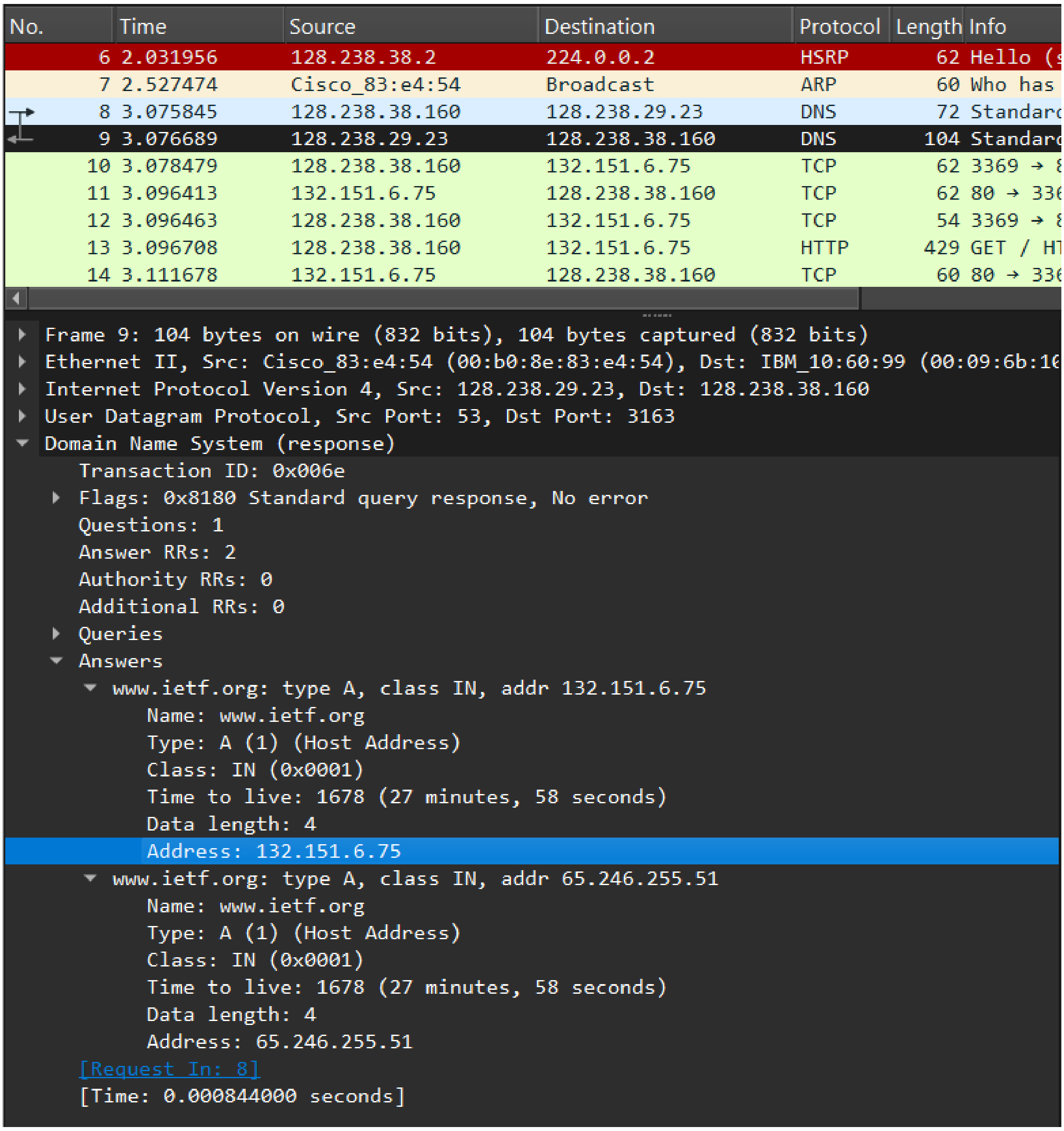


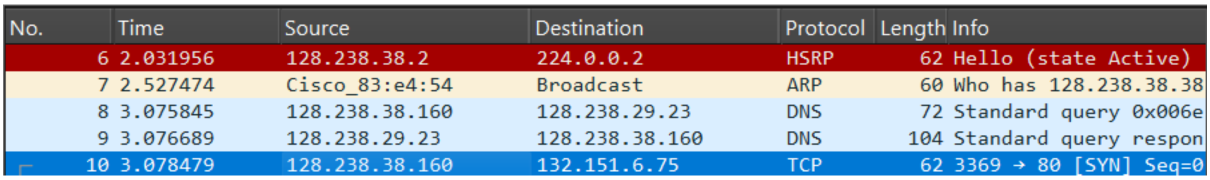
- Content of the 2nd answer:



6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

   - The destination IP address of the SYN packet correspond to one of the IP addresses provided in the DNS response message which is 132.151.6.75.





7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

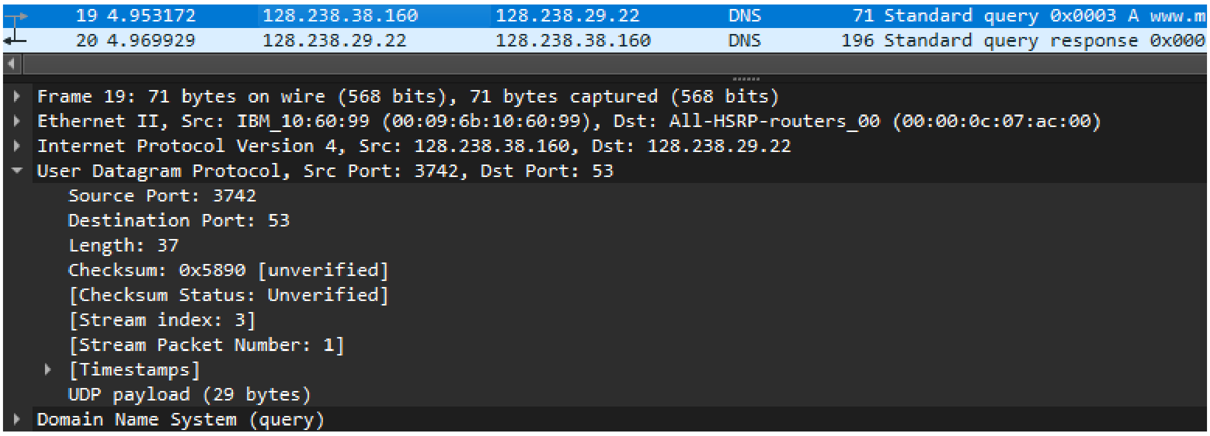
   - No, there are no DNS queries from host before retrieving each image.

No.	Time	Source	Destination	Protocol	Length	Info
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK]
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=1
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1
16	3.128093	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=3
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=2
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=3
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055	HTTP/1.1 200 OK (tex
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=3
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK]
23	3.174716	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=5
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0
25	3.179283	128.238.38.160	132.151.6.75	TCP	62	3371 → 80 [SYN] Seq=0
26	3.191649	132.151.6.75	128.238.38.160	TCP	62	80 → 3370 [SYN, ACK]
27	3.191726	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=1
28	3.191998	128.238.38.160	132.151.6.75	HTTP	320	GET /images/ietflogo2
29	3.192665	132.151.6.75	128.238.38.160	TCP	62	80 → 3371 [SYN, ACK]

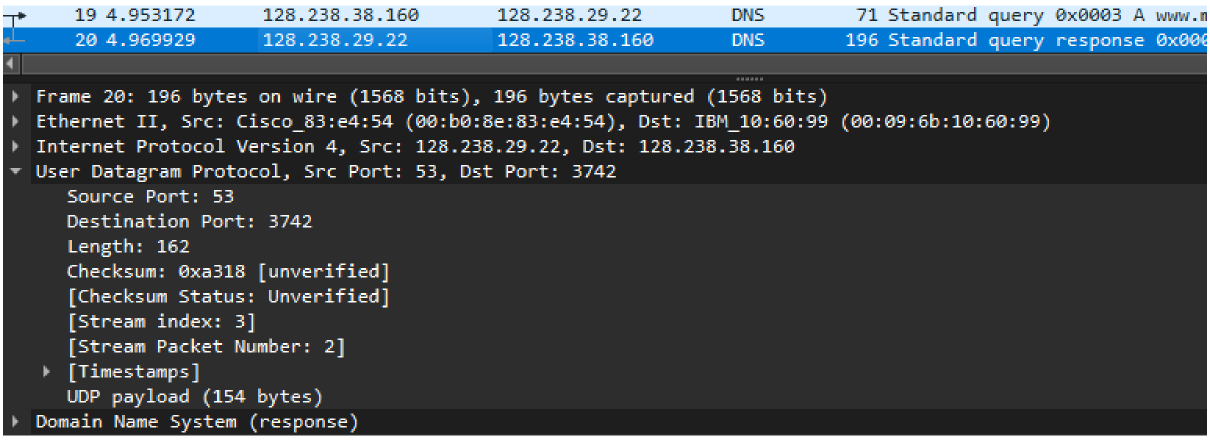
- Open packet trace file dns-trace-2 for nslookup.
- We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
- Answer the following questions.

8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

    - The destination port for the DNS query message is 53.

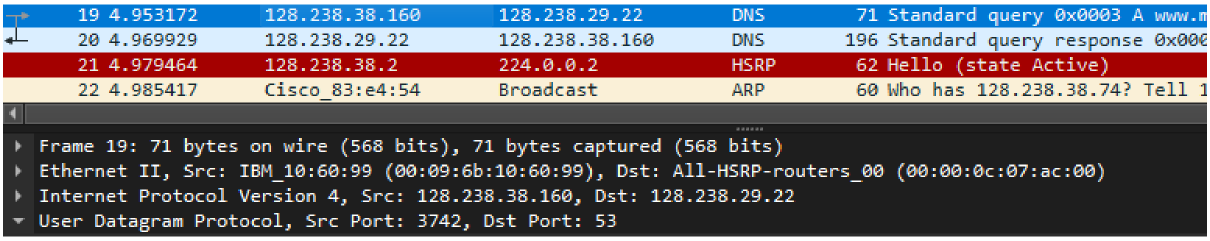


    - The source port of DNS response message is 53.



9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.

- The DNS query message is sent to an IP address of 128.238.29.22.



19	4.953172	128.238.38.160	128.238.29.22	DNS	71 Standard query 0x0003 A www.m
20	4.969929	128.238.29.22	128.238.38.160	DNS	196 Standard query response 0x000
21	4.979464	128.238.38.2	224.0.0.2	HSRP	62 Hello (state Active)
22	4.985417	Cisco_83:e4:54	Broadcast	ARP	60 Who has 128.238.38.74? Tell 1

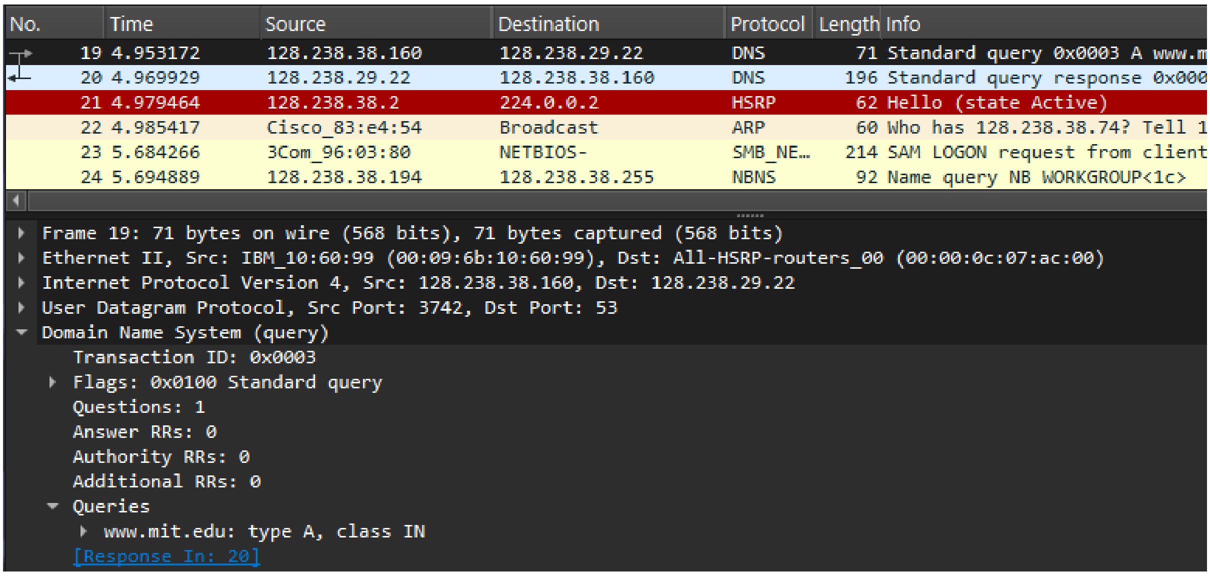
```
Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits)
Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22
User Datagram Protocol, Src Port: 3742, Dst Port: 53
```

- No, it is not the IP address of my default local DNS server.

```
DNS Servers . . . . . . . . . . . : 2001:e68::b:68
                                    2001:e68::b:69
                                    192.168.0.1
                                    2001:e68::b:68
                                    2001:e68::b:69
```

10. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"? Add screenshots in your answer.

- Type A
- There are no answers

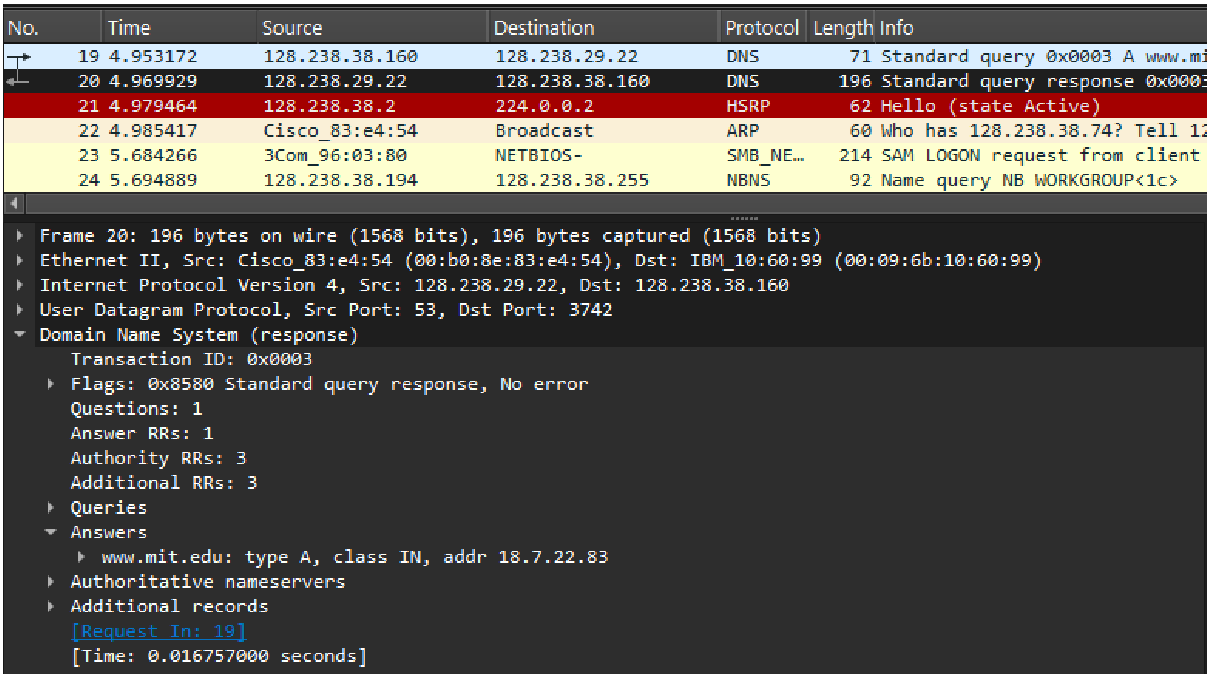


No.	Time	Source	Destination	Protocol	Length	Info
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.m
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x000
21	4.979464	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)
22	4.985417	Cisco_83:e4:54	Broadcast	ARP	60	Who has 128.238.38.74? Tell 1
23	5.684266	3Com_96:03:80	NETBIOS-	SMB_NE…	214	SAM LOGON request from client
24	5.694889	128.238.38.194	128.238.38.255	NBNS	92	Name query NB WORKGROUP<1c>

```
Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits)
Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22
User Datagram Protocol, Src Port: 3742, Dst Port: 53
Domain Name System (query)
    Transaction ID: 0x0003
    Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    Queries
        www.mit.edu: type A, class IN
    [Response In: 20]
```

11. Examine the DNS response message. How many "answers" are provided? What do each of these answers contain? Add screenshots in your answer.

- There is one answer



- Answer contents: